

8(a)	Gravitational field is a region of space in which a mass will experience a non-contact gravitational force.	A1
CKW MC	- This definition was assessed in 2017 & 2018 H2 P3. - Refer to DHS “Gravitational Field” notes for reference. - Many candidates confused <i>gravitational field</i> with <i>gravitational field strength</i>. - An <i>area</i> is not the same as a <i>volume</i> or <i>region</i>. - For definitions, avoid using the root words again. Thus, candidates should explain the words <i>gravitational</i> and <i>field</i> separately.	
8(b)(i)	$\rho = \frac{M}{V} = \frac{M}{\left(\frac{4}{3}\pi r^3\right)} = \frac{0.73}{\left(\frac{4}{3}\pi(25 \times 10^{-3})^3\right)}$ $= 11000 \text{ kg m}^{-3}$	C1 A1
CKW MC	There was quite a number of students who did not use the correct formula for the volume of a sphere or know the importance of the word “diameter”.	
8(b)(ii)	Unit of τ = Unit of $\frac{2\pi^2 mL^2 \theta}{T^2}$ $\text{N m} = \text{kg m s}^{-2} \text{ m} = \frac{\text{kg m}^2}{\text{s}^2}$ Unit of θ Unit of $\theta = 1$ $\tau = \frac{2\pi^2 mL^2 \theta}{T^2}$ $1.2 \times 10^{-5} = \frac{2\pi^2(0.730)(1.80)^2 \theta}{(14.0 \times 60)^2}$ $\theta = 0.18136 \text{ rad}$ $= 0.18136 \left(\frac{360}{2\pi}\right)^\circ = 10(.4)^\circ$	C1 A1 A1
CKW MC	- Most students did not realise the unit of θ in the given equation is in radians. - Some students used $2m$ instead of m.	

8(b)(iii)	Radius of large sphere, $R = D / 2 = 15.0 \text{ cm}$ Radius of small sphere, $r' = d / 2 = 25 \text{ mm} = 2.5 \text{ cm}$ Thus, $r = r' + R + \text{air gap width}$ $= 15.0 + 2.5 + 0.1$ $= 17.6 \text{ cm}$	B1 C1 A0
CKW MC	- Quite a number of students attempted to use Newton's law of gravitation for this part using the given value of torque! - It is also not possible to use the given torque in (b)(ii) for this part since it is a different experiment, and the torque is different for $M = 158 \text{ kg}$ as can be shown that $LF_g = L \left(\frac{GMm}{r^2} \right) = 1.2 \times 10^{-5}$ $r = \sqrt{\frac{LGMm}{1.2 \times 10^{-5}}} = \sqrt{\frac{(1.8)(6.67 \times 10^{-11})(158)(0.73)}{1.2 \times 10^{-5}}} \neq 0.176 \text{ m}$	
8(b)(iv)	Using $s = (L / 2) \theta_1$, $4.1 \times 10^{-3} = (1.8 / 2) \theta_1$ $\theta_1 = 4.5 \times 10^{-3} \text{ rad}$ The uncertainty equation is given by $\frac{\Delta \theta_1}{\theta_1} = \frac{\Delta s}{s} + \frac{\Delta L}{L}$ $\frac{\Delta \theta_1}{4.5 \times 10^{-3}} = \frac{0.1}{4.1} + \frac{0.01}{1.80}$ $\Delta \theta_1 = 0.1 \times 10^{-3} \text{ rad}$ $\theta_1 = (4.6 \pm 0.1) \times 10^{-3} \text{ rad}$	A1 C1 A1
CKW MC	The max-min method will lead to the same answer.	

8(b)(v)	At θ_1, the torque caused by the gravitational attraction (between M and m) was equal to the opposing torque caused by the torsion in the wire. Thus, from (b)(ii), $LF_g = \frac{2\pi^2 mL^2 \theta}{T^2}$ $L \frac{GMm}{r^2} = \left(\frac{2\pi^2 mL^2}{T^2} \right) \theta_1$ $G = \frac{2\pi^2 L r^2 \theta_1}{MT^2}$ (Shown)	B1 C1 A0
	OR The maximum restoring force is equal to the gravitational force between the large and small spheres at a separation of r. Thus, $ F_g = ma = m\omega^2 x_o$ $\frac{GMm}{r^2} = m \left(\frac{2\pi}{T} \right)^2 x_o$ $G = \left(\frac{4\pi^2}{T^2} \right) \frac{r^2}{M} x_o$ The amplitude x_o of the simple harmonic motion is approximately equal to $\frac{L}{2} \theta_1$. Thus, $G = \left(\frac{4\pi^2}{T^2} \right) \frac{r^2}{M} \left(\frac{L}{2} \theta_1 \right)$ $G = \frac{2\pi^2 L r^2 \theta_1}{MT^2}$ (Shown)	B1 C1 A0

CKW MC	Many candidates seem to have the misconception that gravitational force <i>is</i> the torque and/or the squared term for L gets omitted or incorrectly simplified in one step e.g. $\frac{GMm}{r^2} = \left(\frac{2\pi^2 mL^2}{T^2} \right) \theta_1$ $G = \frac{2\pi^2 L r^2 \theta_1}{MT^2}$	
8(c)(i)	When the rod rotates by an angle of θ_2, the laser beam deflects by an angle of $2\theta_2$ since the angle of incidence of the laser beam is equal to its angle of reflection. Thus, Using $s_2 = D_2(2\theta_2)$, $15.6 / 100 = (12.00)(2\theta_2)$ $\theta_2 = 0.00650 \text{ rad}$ $= 0.00650 \left(\frac{360}{2\pi} \right)^\circ$ $= 0.372^\circ$	C1 A0
	OR Using small angle approximation, $2\theta_2 \approx \frac{s_2}{D_2} \rightarrow \tan(2\theta_2) = \sin(2\theta_2) = \frac{s_2}{D_2}$ $\tan(2\theta_2) = \sin(2\theta_2) = \frac{15.6 / 100}{12.00}$ $\theta_2 = 0.372^\circ$	C1 A0
CKW MC	Candidates should show all relevant substitutions and conversion.	

8(c)(ii)	$\frac{3}{4}$ of a period is equal to 10 mins 22 s. Thus, $\frac{3}{4}T = ((10)(60) + 22) \text{ s}$ $T = \frac{2488}{3} \text{ s}$ From (b)(v), $G = \frac{2\pi^2 L r^2 \theta_2}{M T^2}$ $= \frac{2\pi^2 (1.80)(17.6/100)^2 (0.00650)}{(158)(2488/3)^2}$ $= 6.58 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ Unit of G = Unit of $\frac{2\pi^2 L r^2 \theta_2}{M T^2}$ $= \frac{\text{m m}^2}{\text{kg s}^2}$ $= \text{m}^3 \text{ kg}^{-1} \text{ s}^{-2}$	M1 C1 A1 A1
CKW MC	- Most students did not realise the unit of θ_2 is in radians, and conversion needs to be performed for the θ_2 in degrees in (c)(i) if used. - Radian is not an SI base unit.	
8(c)(iii)	T will be determined by taking the average of a large number of oscillations (about 60 instead of about 1 oscillation as in (b)(ii)) with the same actual uncertainty, thus the percentage (or fractional) uncertainty of T will be reduced significantly.	A1
CKW MC	Quite a number of students did not use appropriate and specific terms in their answers such percentage and/or actual uncertainty.	
8(c)(iv)	The rod was assumed to be light.	A1
CKW MC	- Most candidates did not provide the main reason for the difference in T leading to a difference in G. - The system rotates so slowly in 1 oscillation ($T = 14 \text{ min}$) that the damping of air resistance is negligible.	